SG No. 137 D_{4h}^{15} $P4_2/nmc$ [tetragonal]

* plus set: + [0, 0, 0]

Table 1: Wyckoff site: 2a, site symmetry: -4m2

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[1,2,7,8,11,12,13,14]
2	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[3,4,5,6,9,10,15,16]

Table 2: Wyckoff site: 2b, site symmetry: -4m2

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[1,2,7,8,11,12,13,14]
2	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[3,4,5,6,9,10,15,16]

Table 3: Wyckoff site: 4c, site symmetry: 2mm.

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, z\right]$	[1,2,13,14]
2	$\left[\frac{1}{4}, \frac{3}{4}, z+\frac{1}{2}\right]$	[3,4,15,16]
3	$\left[\frac{1}{4}, \frac{3}{4}, -z\right]$	[5,6,9,10]
4	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}-z\right]$	[7,8,11,12]

Table 4: Wyckoff site: 4d, site symmetry: 2mm.

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, z\right]$	[1,2,13,14]
2	$\left[\frac{1}{4}, \frac{1}{4}, z+\frac{1}{2}\right]$	[3,4,15,16]
3	$\left[\frac{3}{4}, \frac{3}{4}, -z\right]$	[5,6,9,10]
4	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}-z\right]$	[7,8,11,12]

Table 5: Wyckoff site: 8e, site symmetry: -1

No.	position	mapping
1	[0, 0, 0]	[1,9]
2	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	[2,10]
3	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	[3,11]
4	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	[4,12]
5	$\left[0, \frac{1}{2}, 0\right]$	[5,13]
6	$\left[\frac{1}{2}, 0, 0\right]$	[6,14]
7	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[7,15]
8	$\left[0, 0, \frac{1}{2}\right]$	[8,16]

Table 6: Wyckoff site: `8f`, site symmetry: `..2`

No.	position	mapping
1	$\left[x,\ -x,\ \frac{1}{4}\right]$	`[1,8]`
2	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ \frac{1}{4}\right]$	`[2,7]`
3	$\left[x+\frac{1}{2},\ x,\ \frac{3}{4}\right]$	`[3,6]`
4	$\left[-x,\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	`[4,5]`
5	$\left[-x,\ x,\ \frac{3}{4}\right]$	`[9,16]`
6	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	`[10,15]`
7	$\left[\frac{1}{2}-x,\ -x,\ \frac{1}{4}\right]$	`[11,14]`
8	$\left[x,\ x+\frac{1}{2},\ \frac{1}{4}\right]$	`[12,13]`

Table 7: Wyckoff site: `8g`, site symmetry: `.m.`

No.	position	mapping
1	$\left[\frac{1}{4},\ y,\ z\right]$	`[1,14]`
2	$\left[\frac{1}{4},\ \frac{1}{2}-y,\ z\right]$	`[2,13]`
3	$\left[\frac{1}{2}-y,\ \frac{1}{4},\ z+\frac{1}{2}\right]$	`[3,15]`
4	$\left[y,\ \frac{1}{4},\ z+\frac{1}{2}\right]$	`[4,16]`
5	$\left[\frac{3}{4},\ y+\frac{1}{2},\ -z\right]$	`[5,10]`
6	$\left[\frac{3}{4},\ -y,\ -z\right]$	`[6,9]`
7	$\left[y+\frac{1}{2},\ \frac{3}{4},\ \frac{1}{2}-z\right]$	`[7,11]`
8	$\left[-y,\ \frac{3}{4},\ \frac{1}{2}-z\right]$	`[8,12]`

Table 8: Wyckoff site: `16h`, site symmetry: `1`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1]`
2	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z\right]$	`[2]`
3	$\left[\frac{1}{2}-y,\ x,\ z+\frac{1}{2}\right]$	`[3]`
4	$\left[y,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	`[4]`
5	$\left[-x,\ y+\frac{1}{2},\ -z\right]$	`[5]`
6	$\left[x+\frac{1}{2},\ -y,\ -z\right]$	`[6]`
7	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	`[7]`
8	$\left[-y,\ -x,\ \frac{1}{2}-z\right]$	`[8]`
9	$[-x,\ -y,\ -z]$	`[9]`
10	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ -z\right]$	`[10]`
11	$\left[y+\frac{1}{2},\ -x,\ \frac{1}{2}-z\right]$	`[11]`
12	$\left[-y,\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	`[12]`
13	$\left[x,\ \frac{1}{2}-y,\ z\right]$	`[13]`
14	$\left[\frac{1}{2}-x,\ y,\ z\right]$	`[14]`
15	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	`[15]`
16	$\left[y,\ x,\ z+\frac{1}{2}\right]$	`[16]`